

HARDIK SRIVASTAVA

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[Portfolio](#) | [LinkedIn](#) | [LeetCode](#) | [ResearchGate](#) | [Google Scholar](#) | [GitHub](#)

EDUCATION

University of Washington <i>Master of Science in Data Science</i>	Seattle, WA Sept 2025 - Mar 2027
SRM Institute of Science and Technology <i>Bachelor of Technology in Computer Science with specialization in Big Data Analytics (GPA - 3.82/4)</i>	Chennai, TN May 2019 - Jun 2023

WORK EXPERIENCE

Graduate Research Assistant (P.I. - Prof. Zaid Harchaoui) Paul G. Allen School of Computer Science, University of Washington	Sept 2025 - Present Seattle, WA
• Enhanced user privacy through authorship obfuscation by fine-tuning modular LoRA adapters for controllable text style transfer; built metric-driven agents to guide this via learned style embeddings and constrained decoding under semantic constraints. • Formulated closed-loop control logic to dynamically remix text across 16+ writing styles, integrating Mistral models for automated evaluation. Boosted author anonymity by 40% on a 30K+ corpus while preserving 88% semantic similarity and 5% perplexity drift. • Optimized agent policies via multi-objective feedback and conducted empirical analysis across multilingual datasets. Benchmarked the system against larger LLM baselines, achieving privacy gains at 3x lower inference cost and 34% fewer generated tokens.	

Applied Scientist 2 JPMorgan Chase & Co.	Jun 2023 - Aug 2025 Hyderabad, TS
• Led fraud detection initiatives saving over \$220M annually by developing high-recall XGBoost and CatBoost; optimized recall-precision tradeoffs using constrained objective tuning while maintaining feature interpretability for stakeholders. • Productionized Semi-Supervised Graph Neural Networks across 60M+ transaction nodes; introduced topology-aware edge sampling and dynamic neighborhood aggregation, boosting fraud-detection precision by 9.2% over 2M transactions/day. • Architected a GPU-served contrastive pre-trained Transformer (ViT + BERT-style decoder), scaling to 500K+ requests/day, for OCR-based check fraud detection. Improved fraud precision by 19% while reducing latency via batched mixed-precision inference. • Built a Learning-to-Rank entity matching system over 3B+ KYC records using Pairwise Ranking loss to flag fraud entities; Outperformed the existing TF-IDF baselines, cutting 62% false positives and saving 700+ analyst review hours daily in production. • Developed distributed data pipelines in PySpark to handle terabyte-scale transactions. Performed optimal feature selection using Decision Trees that boosted Fraud Capture Rate by 14% and enhanced predictive accuracy for risk scoring models.	

Research Intern McGill University - <i>Mitacs Scholar</i>	Jun 2022 - Sept 2022 Montreal, QC
• Developed RoBERTa-based sentence embeddings on AAC image captions to capture word-context. Achieved Krippendorff's alpha reliability score of 0.82 for context-aware vocab retrieval; deployed the model in ClickAAC mobile application.	

Machine Learning Engineer - Intern Samsung Research	Jan 2022 - Jun 2022 Bangalore, KN
• Optimized word-context relationships for Samsung Bixby search using bipartite graphs, reducing false-positive preds by 12%. Matched FP32 perf for INT8 inference using quantization-aware training for word-disambiguation task with a 2.9% F1 gain.	

PROJECTS

Agentic Ad Intelligence Engine [\[code\]](#) Architected a **multi-agent orchestration** pipeline for X (Twitter) using **Grok 4.1** Reasoning models to generate dynamic user personas and personalized ad copy from real-time behavioral signals. Engineered an LLM-driven routing system that uses ensemble engagement predictions and feedback-driven ranking to optimize context-aware ad delivery and in-feed placement.

KeyCognition: Behavioral AI Platform [\[code\]](#) Engineered a temporal sequence modeling pipeline in Databricks, processing **136M+** real-time **keystroke** events to construct latent behavioral embeddings for **cognitive load prediction**. Developed a hybrid behavioral classification model fusing LSTMs and K-Means clustering to extract micro-temporal typing anomalies like backspaces, flight times etc.

PUBLICATIONS

• Srivastava, H. (2023). Multi-Modal Sentiment Analysis using Text and Audio for Customer Support Centers • Srivastava, H. (2023). Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems • Srivastava, H. (2021). Using NLP Techniques for Enhancing Augmentative and Alternative Communication Applications	
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SKILLS

AI/Research	Machine Learning, Natural Language Processing, Generative AI, Graph-ML, Transformers, Ranking, Agentic AI
	Large Language Models, Time Series Forecasting, Data Analysis
CS/Tooling	C++, Python, Java, SQL, PyTorch, TensorFlow, Sklearn, Spark, Kafka, W&B, HuggingFace, Optuna, Git
Infra/MLOps	AWS, Docker, Kubernetes, Flask, MLFlow, RAG, LangChain, NVIDIA Triton, Feature Stores, Jenkins